

Public Companies: Critical Minerals for AI, Defense & Advanced Manufacturing

Rare earths, copper, and constrained inputs where the US faces meaningful supply/China-dependency risk

1. Rare Earth Elements & Permanent Magnets

Neodymium, dysprosium, and related elements used in magnets for actuator motors, EVs, wind turbines, and defense systems. China controls ~90-95% of global processing.

Company	Ticker	Role	Recent Signal (2026)
MP Materials	MP	Operates Mountain Pass, the only active scale rare earth mine/processor in the US	\$400M DoD equity investment and 10-year offtake agreement with price floors (2025); added to China's export control list (June 2026)
USA Rare Earth	USAR	Building sintered NdFeB magnet plant in Oklahoma; developing heavy rare earth supply chain	LOI for \$1.6B in CHIPS Act funding plus \$1.5B private capital (Jan 2026); also added to China's export control list
Lynas Rare Earths	ASX: LYC	Largest rare earth producer outside China; Mt Weld mine (Australia) and Malaysia processing	Central beneficiary of allied-nation rare earth diversification push amid tight global supply (2026)
Ucore Rare Metals	TSXV: UCU	Rare earth separation technology (RapidSX) for North American supply chain	Advancing US-based separation facility development

2. Copper (AI Data Center & Grid Demand Driver)

Not classically a 'critical mineral' but structurally short: a single large AI data center can require up to 50,000 metric tons; J.P. Morgan projects a 2026 refined copper shortfall of roughly 330,000 metric tons.

Company	Ticker	Role	Recent Signal (2026)
Freeport-McMoRan	FCX	Largest US-listed copper producer	Direct beneficiary of AI/data-center copper demand thesis and structural supply deficit
Southern Copper	SCCO	Major low-cost copper producer (Peru/Mexico)	Benefiting from copper prices holding near \$4.80-5.10/lb through 2026
Ivanhoe Mines	TSX: IVN	Kamoa-Kakula, one of the world's largest copper mines (DRC)	Highlighted as top Canadian copper stock leveraged to global copper demand growth (2026)
Teck Resources	TSX: T ECK.B	Diversified Canadian miner with growing copper production	331,232 tonnes produced in 2025; shares up 31% over trailing year
Rio Tinto	RIO	Global diversified miner with major copper assets (Arizona, Kennecott)	Signed 2-year agreement to supply AWS with domestically produced Arizona copper (Jan 2026)
Hudbay Minerals	TSX: HBM	Copper/zinc producer with North and South American assets	~\$4.9B market cap, benefiting from copper supply-deficit repricing (2026)

3. Tungsten

Critical for semiconductor conductive layers, high-precision machining, and munitions. China's 2025 export ban left US refining infrastructure (built in the 1950s) running far below its ~18,000-tonne capacity due to feedstock shortage.

Company	Ticker	Role	Recent Signal (2026)
Almonty Industries	ALM / TSX: All	Largest tungsten producer outside China; Sangdong mine (South Korea), Panasqueira (Portugal)	Q1 2026 revenue up ~221% YoY on rising tungsten prices; Sangdong Mine began saleable concentrate production (2026); \$700M convertible notes offering

4. Antimony

Critical for munitions (tracer rounds, incendiaries), flame retardants, and optical systems. China's August 2024 export restrictions made this an immediate national security concern.

Company	Ticker	Role	Recent Signal (2026)
United States Antimony Corp	UAMY	Only fully integrated antimony producer/processor outside China and Russia; also produces zeolite, cobalt, tungsten	Continues to expand critical-minerals processing footprint as of 2026 ("The Critical Minerals and ZEO Company")
Perpetua Resources	PPTA	Developing Idaho antimony-gold project, a rare domestic antimony source	Advancing toward production as US seeks non-China antimony supply

5. Gallium & Germanium (Semiconductor-Critical)

Gallium underpins GaN power electronics and RF chips; germanium is essential to optical fiber. Both were subject to China export bans (partially, temporarily suspended through Nov 27, 2026). Supply is structurally tight because both are byproducts of bauxite/zinc processing, not independently mined.

Company	Ticker	Role	Recent Signal (2026)
Sunshine Silver Mining & Refining	Recent S-1/IP O filer	Silver miner with gallium/germanium byproduct exposure disclosed in 2026 IPO filings	Filed S-1/424B4 in 2026 explicitly citing gallium/germanium demand growth tied to semiconductors and data centers

Note: there is no pure-play US public gallium or germanium producer -- both are byproducts of other mining operations, which is itself the core supply vulnerability. This is a thinner, higher-risk part of the list than the other categories.

Categories Worth Adding

- **Battery & Energy Transition Metals** (lithium, cobalt, nickel, graphite) — critical for grid-scale storage supporting AI power buildout, but the investment thesis is more EV-driven than AI-specific; worth a separate pass if you want full coverage.
- **Molybdenum** — gaining relevance as chipmakers transition from tungsten to molybdenum in next-gen memory chips; currently a mostly China-listed/private theme with limited US pure-plays.
- **Critical Minerals Recycling** — e-waste/urban mining recovery of gallium, germanium, and rare earths from discarded electronics; a policy-driven category (e.g. proposed Washington State "Semiconductor Stewardship Act") rather than an established public-market segment yet.

- **Allied-Nation Supply Chain Vehicles** — joint ventures and offtake structures outside pure miners (e.g. International Development Finance Corporation-backed consortiums); not directly investable as single stocks but shape which miners get funded.

Generated July 10, 2026. This is a disjoint reference set, not a ranked or filtered list.